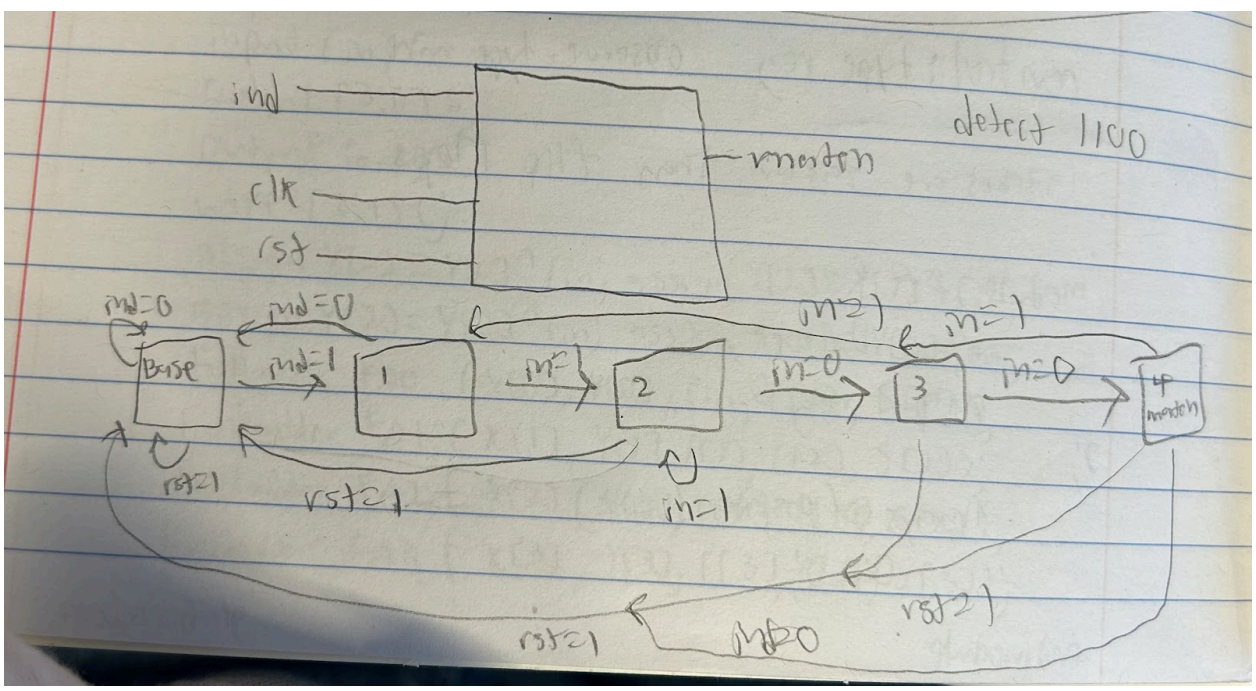
Dylan Thornburg ECEN 127 HW 3

In class I wrote a state machine that detects an input sequence 101. I want you to develop a similar sequence detector. Your detector should output a 1 when the prior 4 inputs are 1100 in that order and output a 0 in all other cases. All inputs and outputs are active high and the reset is synchronous.

Write a Verilog module and a testbench. Turn these in with screenshots or other output that indicates that it works for several test cases. Also provide a state diagram.

STATE DIAGRAM



SM Code

```
module sm(
	input clk,
	input rst,
	input ind,
	output reg match
);
	parameter StateBASE = 3'b000;
	parameter State1   = 3'b001;
	parameter State2   = 3'b010;
	parameter State3   = 3'b011;
	parameter State4   = 3'b100;

	reg [2:0] curr_state, next_state;

	//detects 1100

	always @(posedge clk) begin
		if (rst) curr_state <= StateBASE;
		else curr_state <= next_state;
	end

	always @(*) begin
		// Default output
		match=1'b0;

		// default next state
		next_state = curr_state;

		case (curr_state)

			//BASE Seq
			StateBASE: begin
				if (ind==1) next_state = State1;
				else next_state = StateBASE;
			end

			//1 Seq
			State1: begin
				if (ind==1) next_state = State2;
				else next_state = StateBASE;
			end

			//2 Seq
			State2: begin
				if (ind==1) next_state = State2;
				else next_state = State3;
			end
```

```
                        //3 Seq
                        State3: begin
                                if (ind==1) next_state = State1;
                                else next_state = State4;
                        end

                        //4 Seq
                        State4: begin
                                match=1'b1;
                                if (ind==1) next_state = State1;
                                else next_state = StateBASE;
                        end

                        //def
                        default: begin
                                next_state = StateBASE;
                        end
                endcase
        end
endmodule
```

TB Code

```
module testbench();

reg clk, rst, ind;
wire match;

sm dut(clk, rst, ind, match);

initial clk=1;
always #5 clk = ~clk;


initial begin
$display("Start of the state machine test. Remember to ignore clk=0 lines.");
$monitor("indata=%d, rst=%d, match=%d, clk=%d @%d", ind, rst, match, clk, $time);
ind=1'b0; rst=1'b1;
#10
$display("Reset finished to ensure no problems");
ind=1'b1; rst=1'b0;
#10
ind=1'b1; rst=1'b0;
#10
ind=1'b0; rst=1'b0;
```

```
#10
ind=1'b0; rst=1'b0;
#10
$display("match should be one ^^^");
ind=1'b0; rst=1'b0;
#10
ind=1'b1; rst=1'b0;
#10
ind=1'b1; rst=1'b0;
#10
ind=1'b0; rst=1'b1;
#10
ind=1'b1; rst=1'b0;
#10
ind=1'b1; rst=1'b0;
#10
ind=1'b1; rst=1'b0;
#10
ind=1'b0; rst=1'b0;
#10
ind=1'b0; rst=1'b0;
#10
$display("match should be one ^^^");
ind=1'b0; rst=1'b0;
#10
ind=1'b0; rst=1'b0;
#10
ind=1'b1; rst=1'b0;
#10
ind=1'b1; rst=1'b0;
#10
ind=1'b0; rst=1'b0;
#10
ind=1'b0; rst=1'b1;
#10
$display("match should not be one, as we reset even though we got the 1100 input ^^^");
$display("End of SM Test");
$finish;

end

endmodule
```

RESULTS

```
[dthornburg@linux402604 HW3]$ ./simv -no_save
Chronologic VCS simulator copyright 1991-2023
Contains Synopsys proprietary information.
Compiler version V-2023.12-SP1-1_Full64; Runtime version V-2023.12-SP1-1_Full64;  Oct 16 13:28 2025
Start of the state machine test. Remember to ignore clk=0 lines.
indata=0, rst=1, match=0, clk=1 @                   0
indata=0, rst=1, match=0, clk=0 @                   5
Reset finished to ensure no problems
indata=1, rst=0, match=0, clk=1 @                  10
indata=1, rst=0, match=0, clk=0 @                  15
indata=1, rst=0, match=0, clk=1 @                  20
indata=1, rst=0, match=0, clk=0 @                  25
indata=0, rst=0, match=0, clk=1 @                  30
indata=0, rst=0, match=0, clk=0 @                  35
indata=0, rst=0, match=1, clk=1 @                  40
indata=0, rst=0, match=1, clk=0 @                  45
match should be one ^^^
indata=0, rst=0, match=0, clk=1 @                  50
indata=0, rst=0, match=0, clk=0 @                  55
indata=1, rst=0, match=0, clk=1 @                  60
indata=1, rst=0, match=0, clk=0 @                  65
indata=1, rst=0, match=0, clk=1 @                  70
indata=1, rst=0, match=0, clk=0 @                  75
indata=0, rst=1, match=0, clk=1 @                  80
indata=0, rst=1, match=0, clk=0 @                  85
indata=1, rst=0, match=0, clk=1 @                  90
indata=1, rst=0, match=0, clk=0 @                  95
indata=1, rst=0, match=0, clk=1 @                 100
indata=1, rst=0, match=0, clk=0 @                 105
indata=1, rst=0, match=0, clk=1 @                 110
indata=1, rst=0, match=0, clk=0 @                 115
indata=0, rst=0, match=0, clk=1 @                 120
indata=0, rst=0, match=0, clk=0 @                 125
indata=0, rst=0, match=1, clk=1 @                 130
indata=0, rst=0, match=1, clk=0 @                 135
match should be one ^^^
indata=0, rst=0, match=0, clk=1 @                 140
indata=0, rst=0, match=0, clk=0 @                 145
indata=0, rst=0, match=0, clk=1 @                 150
indata=0, rst=0, match=0, clk=0 @                 155
indata=1, rst=0, match=0, clk=1 @                 160
indata=1, rst=0, match=0, clk=0 @                 165
indata=1, rst=0, match=0, clk=1 @                 170
indata=1, rst=0, match=0, clk=0 @                 175
indata=0, rst=0, match=0, clk=1 @                 180
indata=0, rst=0, match=0, clk=0 @                 185
indata=0, rst=1, match=0, clk=1 @                 190
indata=0, rst=1, match=0, clk=0 @                 195
match should not be one, as we reset even though we got the 1100 input ^^^
End of SM Test
$finish called from file "tb.v", line 60.
$finish at simulation time                 200
           V C S   S i m u l a t i o n   R e p o r t
Time: 200
CPU Time:      0.190 seconds;       Data structure size:   0.0Mb
Thu Oct 16 13:28:48 2025
[dthornburg@linux402604 HW3]$
```